

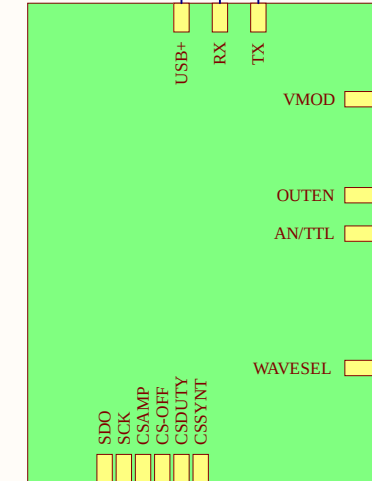
1

2

3

4

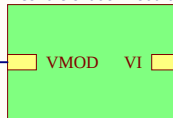
PANEL DE CONTROL
PanelControl.SchDoc



SDO
SCK
CSAMP
CSOFF
CSDUTY
CSSYNT

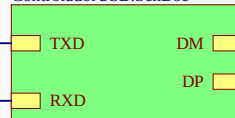
Señal modulante
de 0 a 5v

ACONDICIONADOR
AcondicionadorModulante.SchDoc



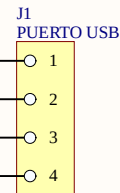
Señal modulante
de 0 a 10v

CONTROLADOR USB
ControladorUSB.SchDoc



R20
1K

R29
10K
GND



J2
ENT. MODULACION

GND

R21
10K
GND

Q1
BC817

D10
1N4148

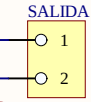
+5v

K1
RELAY-G5V-1

R32
0R

GND

GND



R30
10K
GND

Q3
BC817

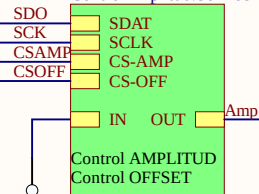
D13
1N4148

+5v

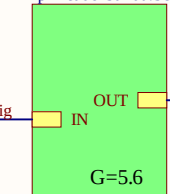
K3
RELAY-G5V-1

Control Amplitud y
offset

AMPLITUD Y OFFSET
ControlAmplitud.SchDoc



AMP. SALIDA
AmplificadorSalida.SchDoc



20vpp

+5v

R22
10K

D11
1N4148

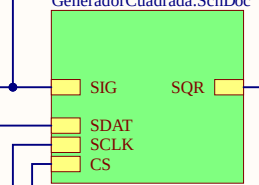
Q2
BC817

GND

Sin+Tri

Square

GEN. CUADRA
GeneradorCuadrada.SchDoc



3.6vpp
1.8vDC

U7A
SN74VLC2G14

6

3

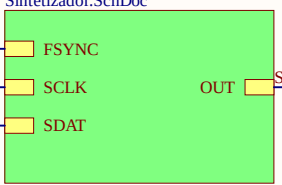
U7B
SN74VLC2G14

4

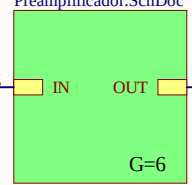
TTL 5v

RELAY-G5V-1

SINTETIZADOR SEN. Y TRIANG.
Sintetizador.SchDoc



PREAMPLIFICADOR
Preamplificador.SchDoc

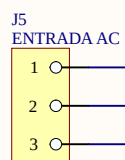


3.6vpp
1.8vDC

SynthSig

Control Frecuencia y
forma de onda

CSSYNT
SCK
SDO

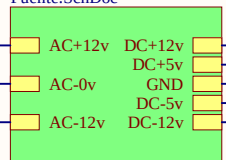


GND

S5

SW DPDT

FUENTE
Fuente.SchDoc



+12v +5v

-12v -5v

GND

600mvpp
300mvDC

Control DUTY

SDO

SCK

CSDUTY

Title **DDS - Diagrama General**

Size: **A4**

Number: **1**

Revision: **2**

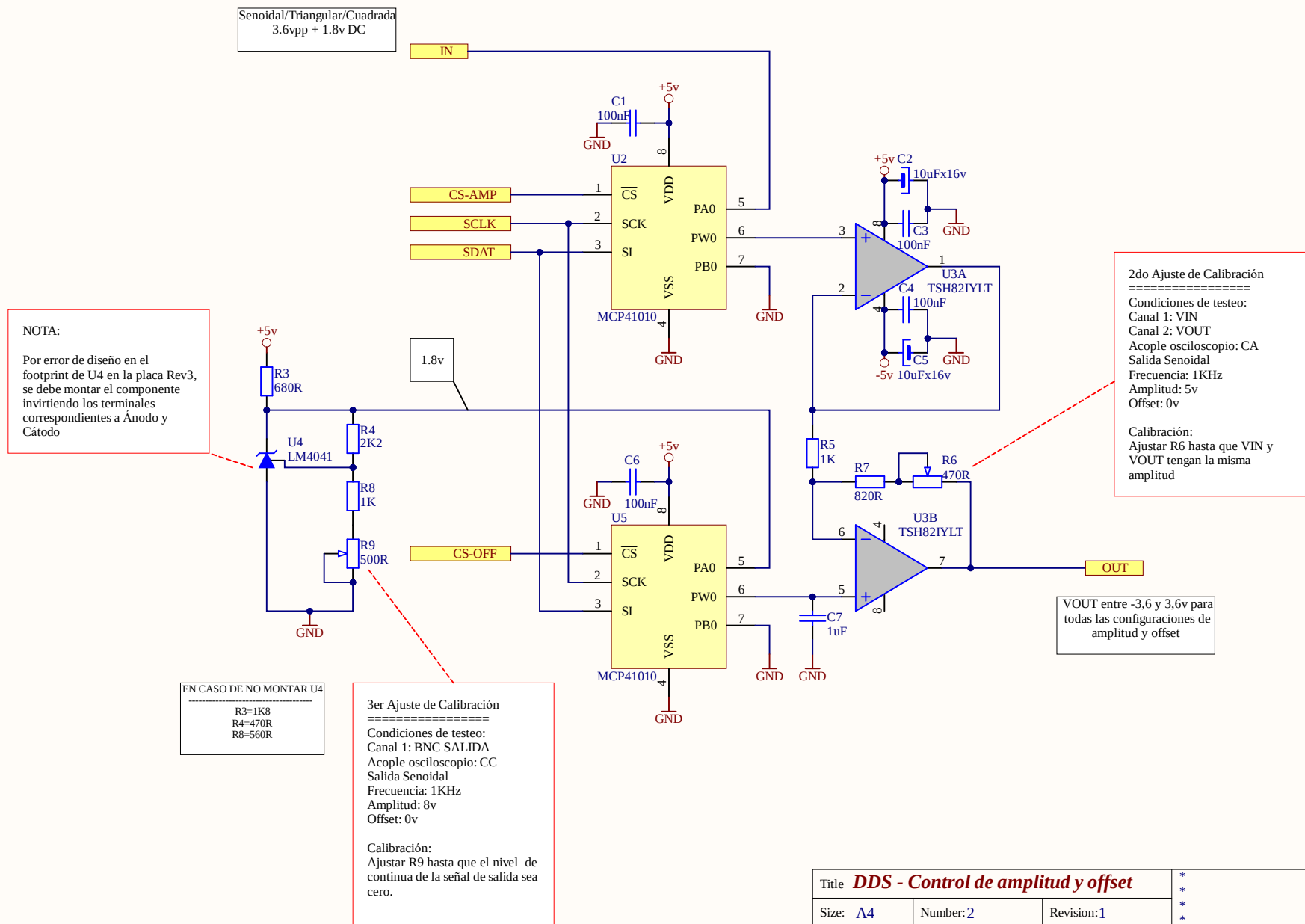
Date: **22/11/2022**

Time: **10:18:34**

Sheet **1** of **10**

File: **D:\Proyectos\DDS\Electronics\DDS Rev2.SchDoc**





Title **DDS - Control de amplitud y offset**

Size: **A4**

Number: **2**

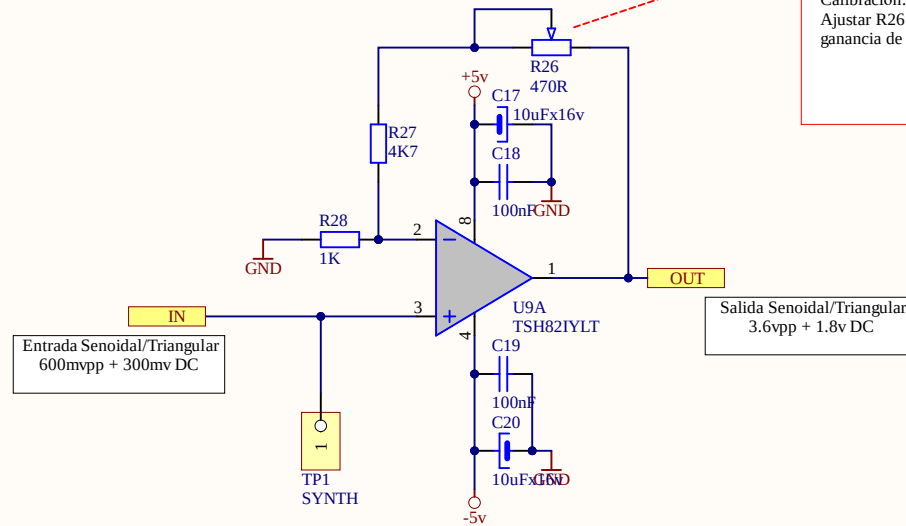
Revision: **1**

Date: **22/11/2022**

Time: **10:18:35**

Sheet **2** of **10**

File: **D:\Proyectos\DDS\Electronics\Control\Amplitud.SchDoc**



1er Ajuste de Calibración

=====

Condiciones de testeo:

Canal 1: VIN

Canal 2: VOUT

Acople osciloscopio: C.A

Salida Senoidal

Frecuencia: 1KHz

Calibración:

Ajustar R26 hasta lograr una ganancia de 6

Entrada Senoidal/Triangular
600mVpp + 300mV DC

Salida Senoidal/Triangular
3.6Vpp + 1.8V DC

Title **DDS - Preamplificador**

Size: **A4**

Number: **3**

Revision: **1**

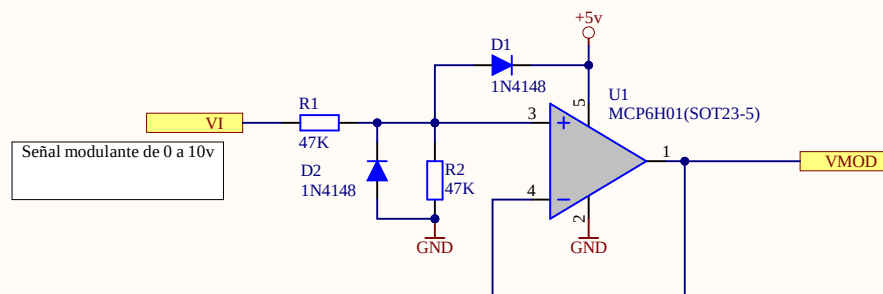
Date: **22/11/2022**

Time: **10:18:35**

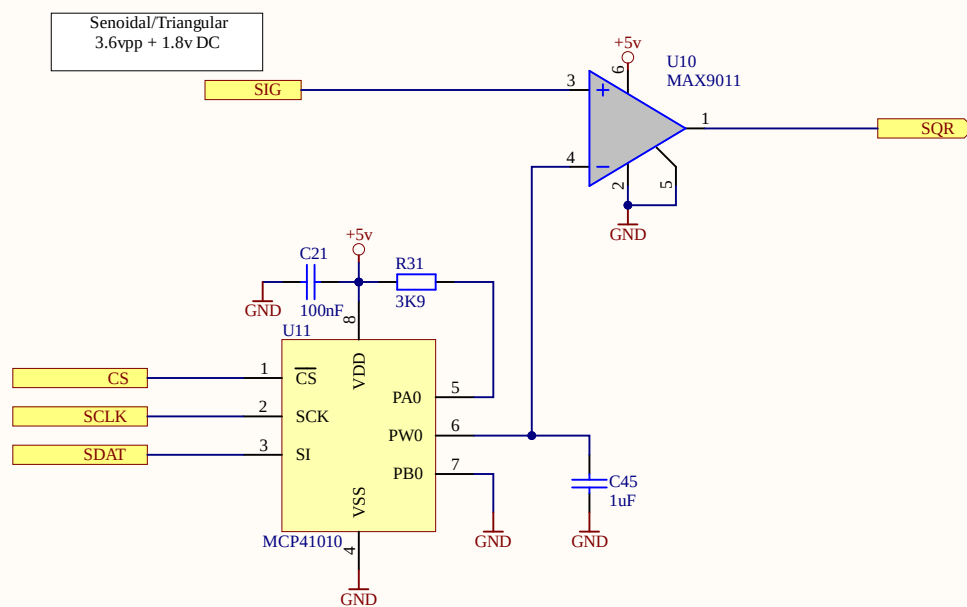
Sheet **3** of **10**

File: **D:\Proyectos\DDS\Electronics\Preamplificador.SchDoc**



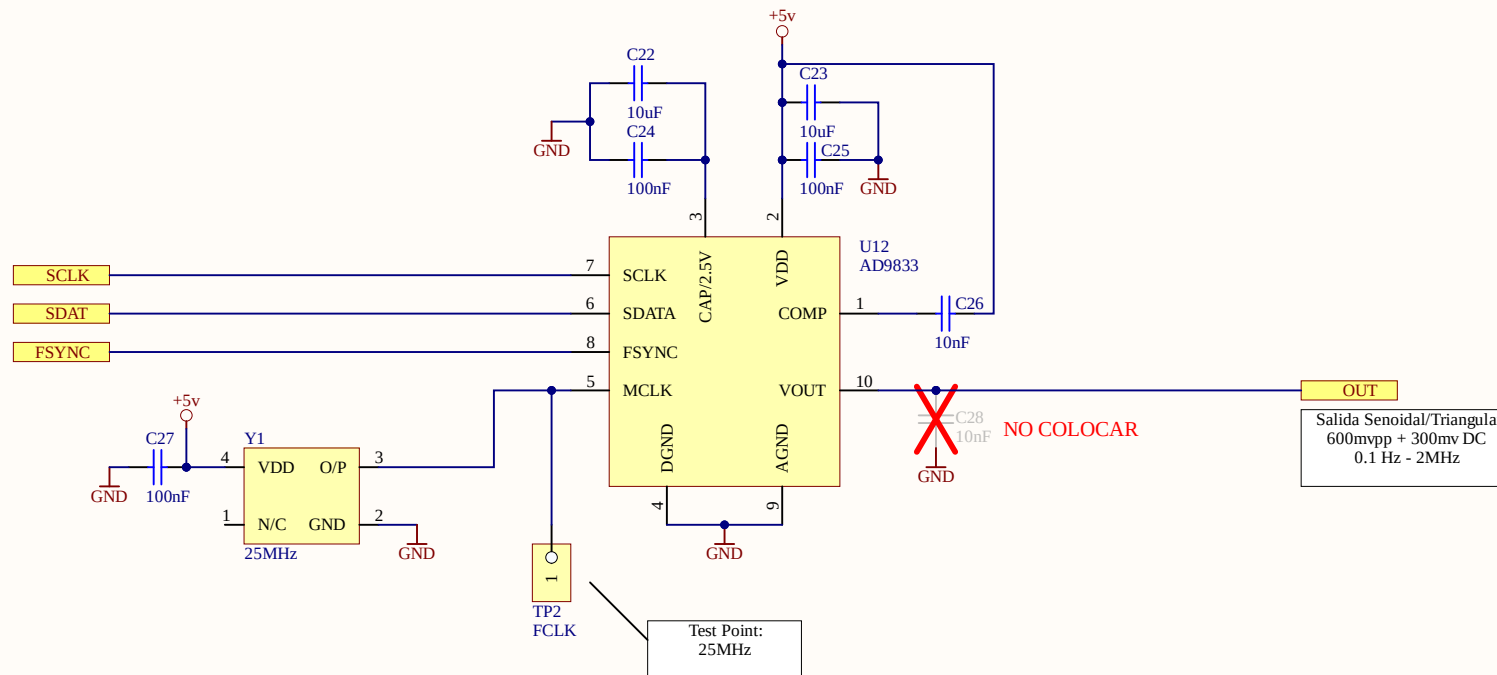


Title DDS - Acondicionador Modulante			* * * *	
Size: A4	Number: 4	Revision: 1		
Date: 22/11/2022	Time: 10:18:35	Sheet 4 of 10		
File: D:\Proyectos\DDS\Electronics\AcondicionadorModulante.SchDoc				



Title DDS - Generador Onda Cuadrada		
Size: A4	Number: 5	Revision: 1
Date: 22/11/2022	Time: 10:18:35	Sheet 5 of 10
File: D:\Proyectos\DDS\Electronics\GeneradorCuadrada.SchDoc		





Title **DDS - Sintetizador Senoidal y Triang.**

Size: **A4**

Number: **6**

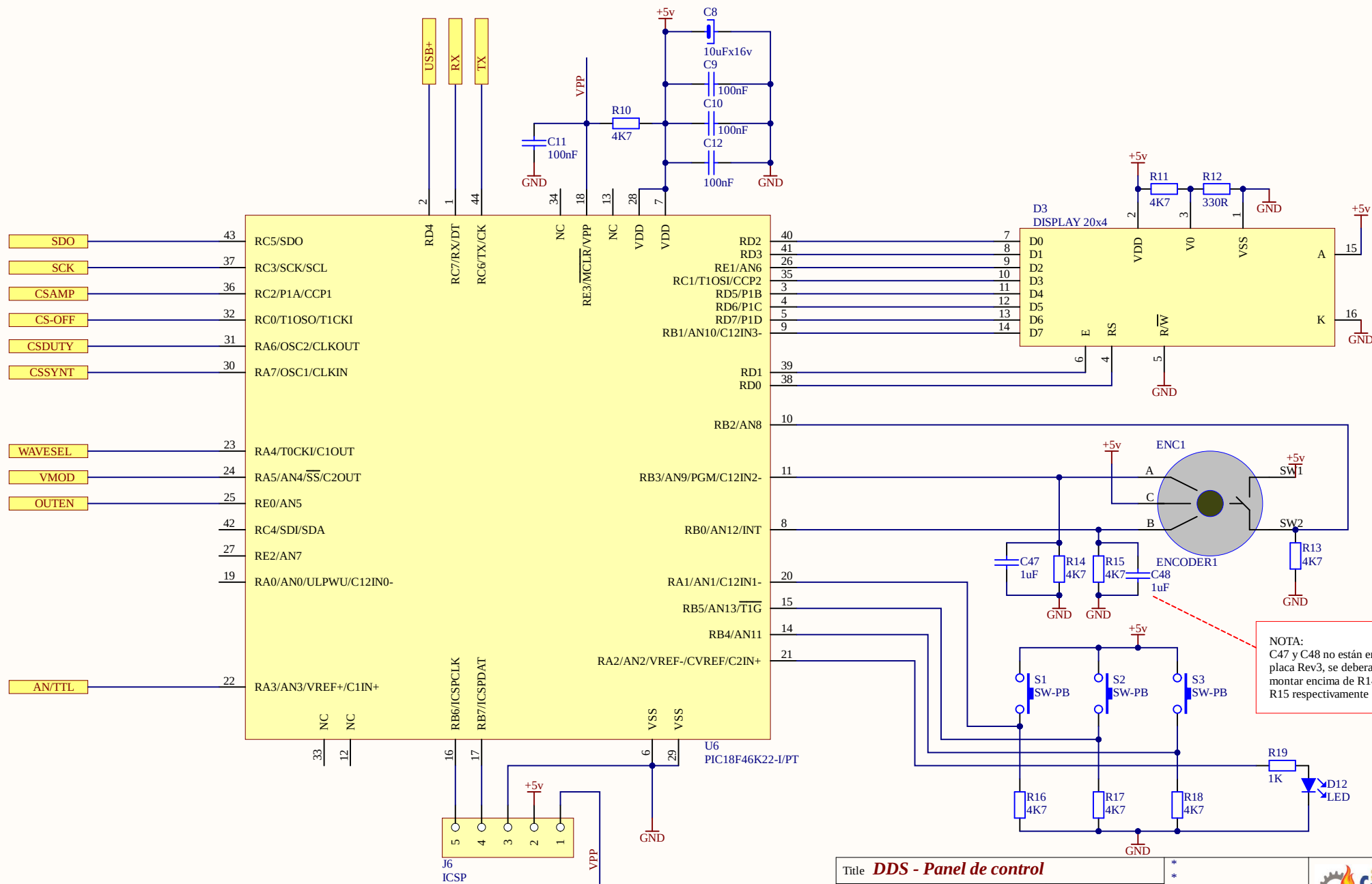
Revision: **1**

Date: **22/11/2022**

Time: **10:18:35**

Sheet **6** of **10**

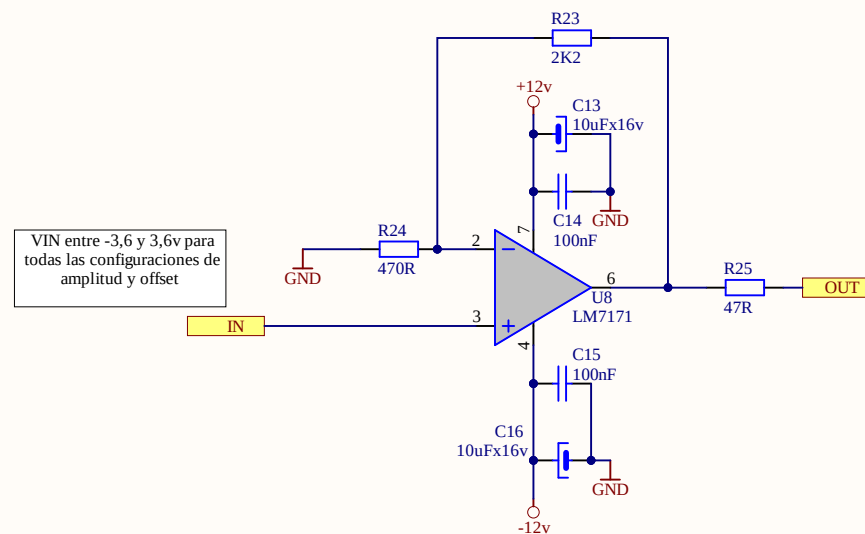
File: **D:\Proyectos\DDS\Electronics\Sintetizador.SchDoc**



NOTA:
C47 y C48 no están en la
placa Rev3, se deberán
montar encima de R14 y
R15 respectivamente

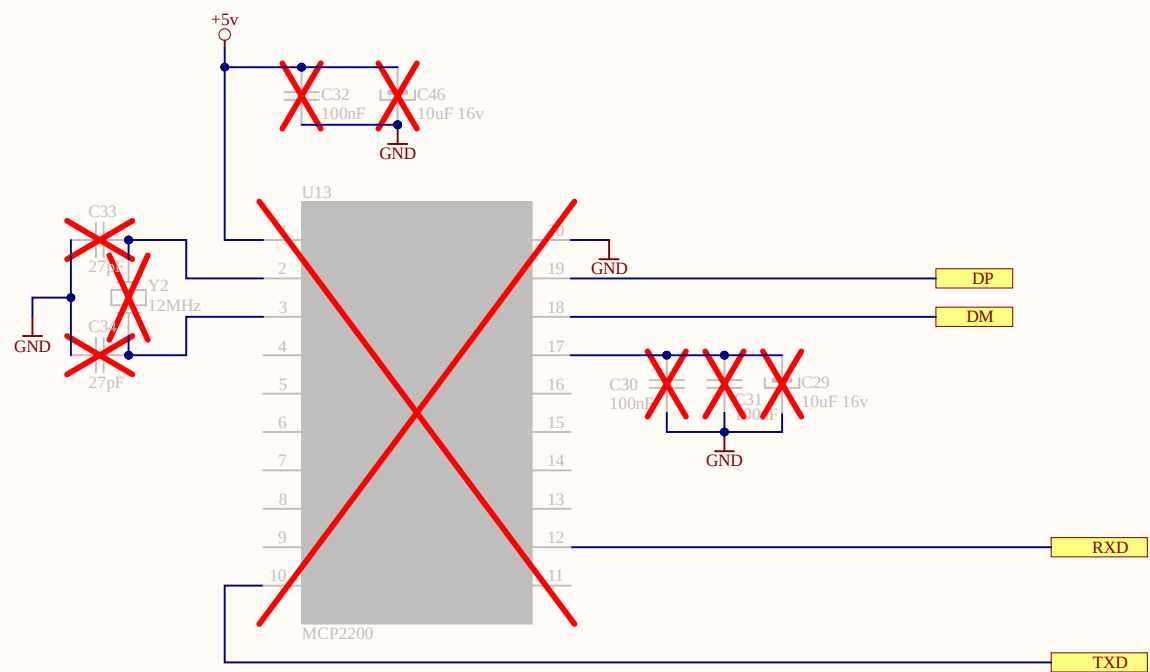
Title DDS - Panel de control		
Size: A4	Number: 7	Revision: 1
Date: 22/11/2022	Time: 10:18:35	Sheet 7 of 10
File: D:\Proyectos\DDS\Electronics\PanelControl.SchDoc		



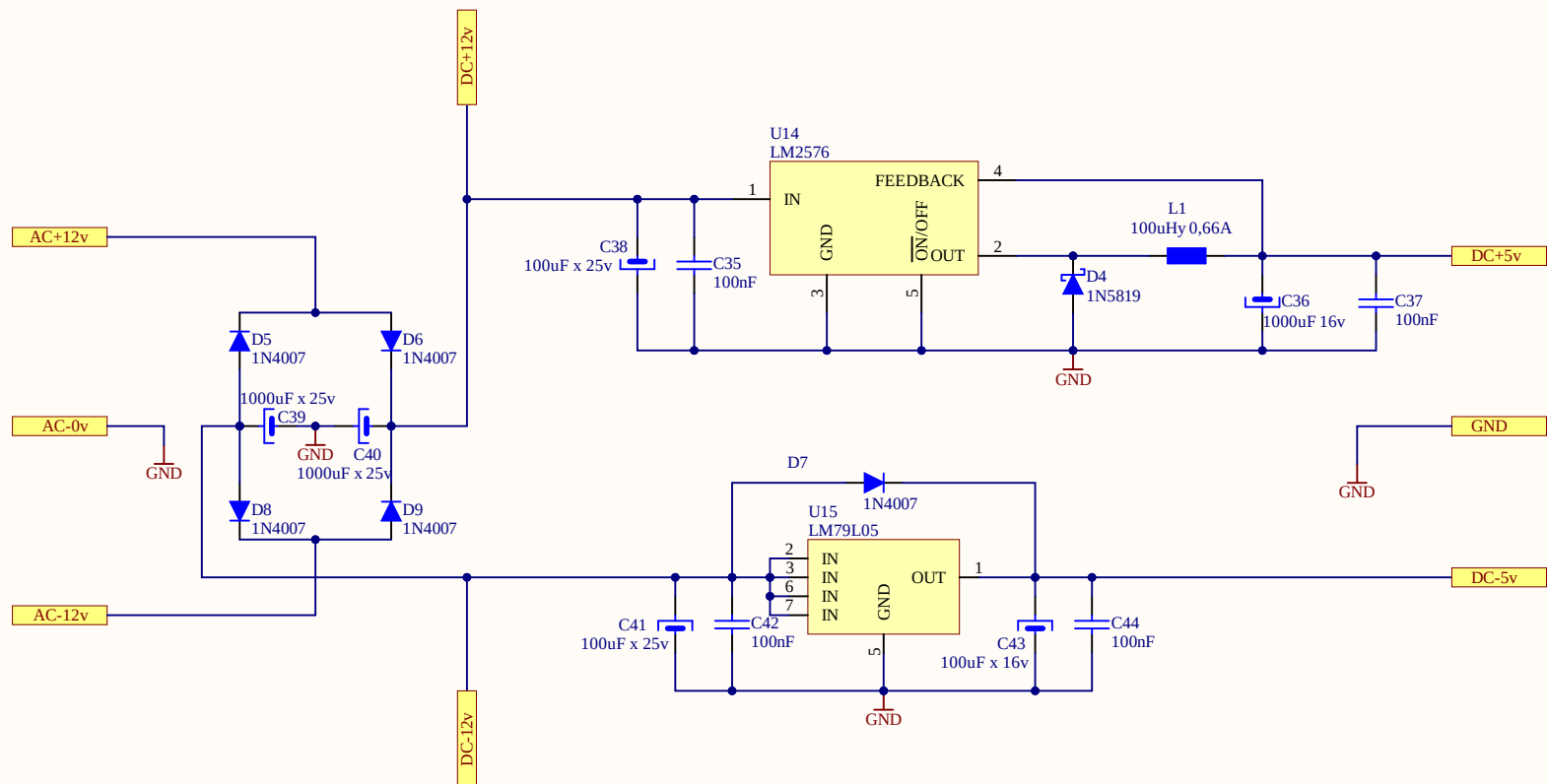


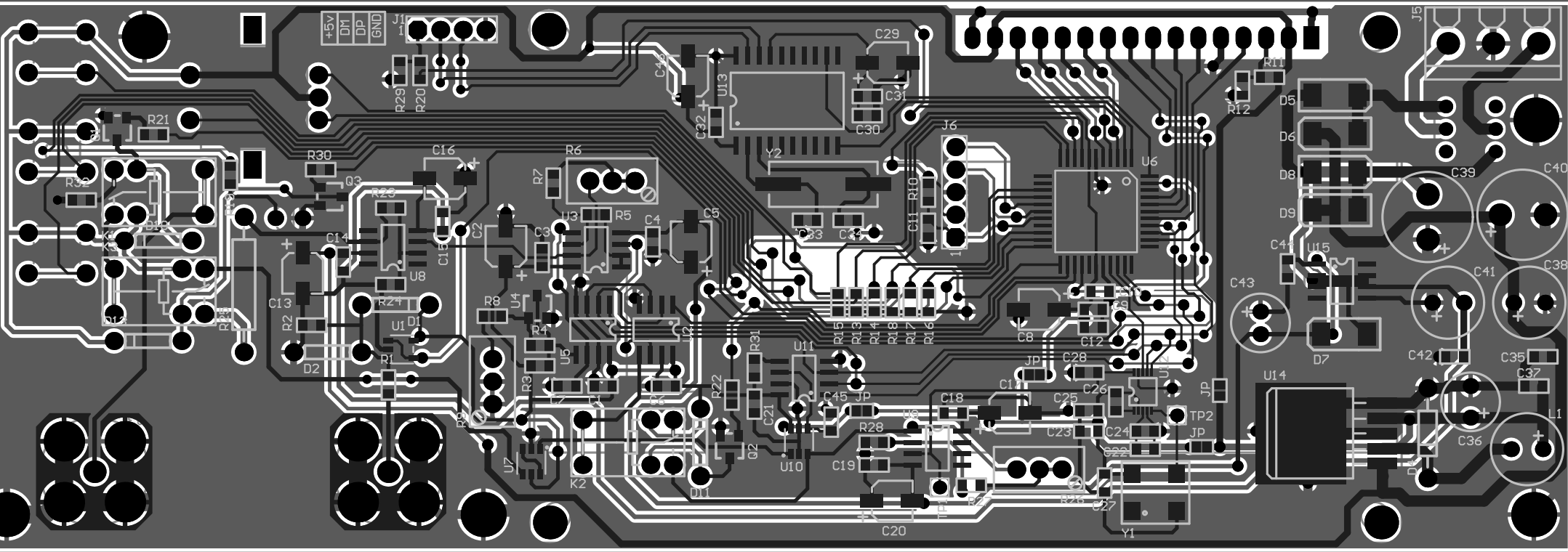
Title DDS - Amplificador de Salida		
Size: A4	Number: 8	Revision: 1
Date: 22/11/2022	Time: 10:18:35	Sheet 8 of 10
File: D:\Proyectos\DDS\Electronics\AmplificadorSalida.SchDoc		



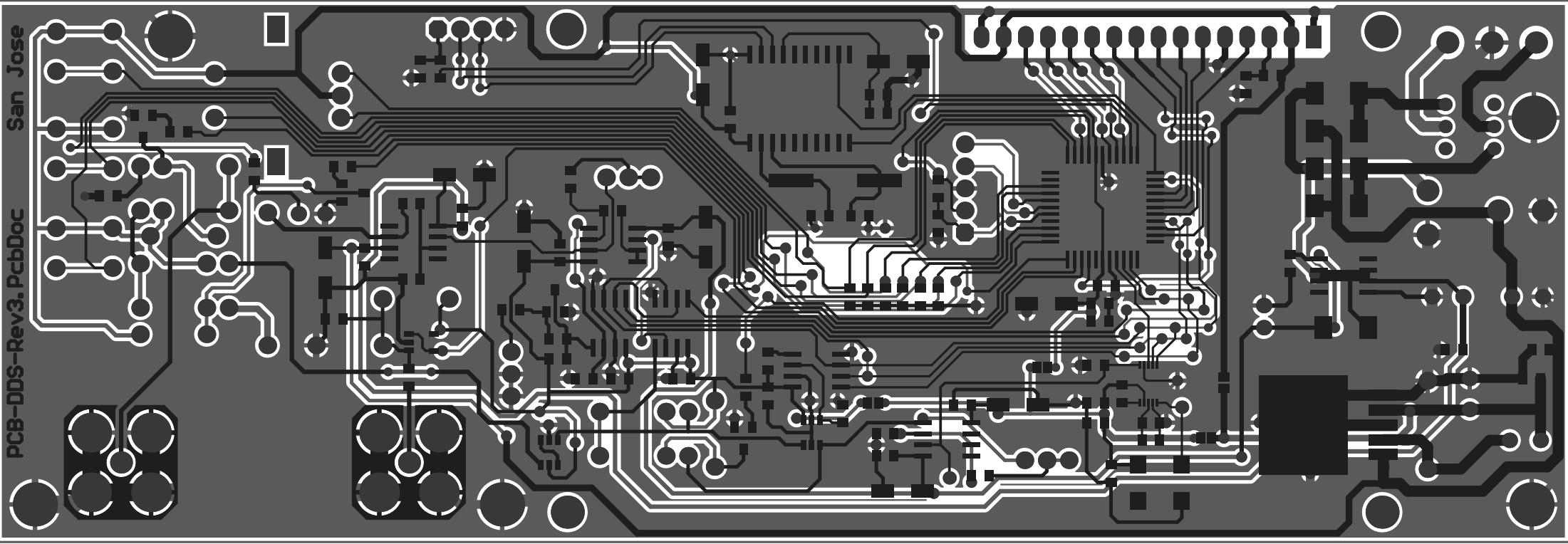


Title DDS - Controlador USB		
Size: A4	Number: 9	Revision: 1
Date: 22/11/2022	Time: 10:18:35	Sheet 9 of 10
File: D:\Proyectos\DDS\Electronics\ControladorUSB.SchDoc		





Quantity	Designator	Comment	Footprint	Description
20	C1, C3, C4, C6, C9, C10, C11, C12, C14, C15, C18, C19, C21, C24, C25, C27, C35, C37, C42, C44	100nF	0805	Capacitor
7	C2, C5, C8, C13, C16, C17, C20	10uFx16v	ALUM_CAP_4mm	Capacitor
4	C7, C45, C47, C48	1uF	0805	Capacitor
2	C22, C23	10uF	0805	Capacitor
1	C26	10nF	0805	Capacitor
1	C36	1000uF 16v	ROUND0.14/0.26	Capacitor
2	C38, C41	100uF x 25v	ROUND0.14/0.32	Capacitor
2	C39, C40	1000uF x 25v	ROUND0.20/0.40	Capacitor
1	C43	100uF x 16v	ROUND0.14/0.26	Capacitor
5	D1, D2, D10, D11, D13	1N4148	DAXIAL0.3	Diode
1	D3	DISPLAY 20x4	TC2004A	Display
1	D4	1N5819	DAXIAL0.4	Diode
5	D5, D6, D7, D8, D9	1N4007	D2010	Diode
1	D12	LED	LED 3mm	
1	ENC1	ENCODER1	ENCODER EN11	
1	J1	PUERTO USB	HDR1x4	Conector
1	J2	ENT. MODULACION	BNC RECT	Conector
1	J3	SALIDA	BNC RECT	Conector
1	J5	ENTRADA AC	BORNERA_3	Conector
1	J6	ICSP	HDR1x5	
3	K1, K2, K3	RELAY-G5V-1	G5V-1	
1	L1	100uHy 0,66A	INDUCTOR0.14/0.26	Inductor
3	Q1, Q2, Q3	BC817	SOT23	Transistor
2	R1, R2	47K	0805	Resistor
1	R3	680R	0805	Resistor
2	R4, R23	2K2	0805	Resistor
5	R5, R8, R19, R20, R28	1K	0805	Resistor
3	R6, R24, R26	470R	PRESET5, 0805	Resistor, Preset
1	R7	820R	0805	Resistor
1	R9	500R	PRESET5	Resistor
9	R10, R11, R13, R14, R15, R16, R17, R18, R27	4K7	0805	Resistor
1	R12	330R	0805	Resistor
4	R21, R22, R29, R30	10K	0805	Resistor
1	R25	47R	AXIAL0.6	Resistor
1	R31	3K9	0805	Resistor
1	R32	0R	0805	Resistor
3	S1, S2, S3	SW-PB	TACT-SWITCH-13mm	Switch
1	S5	SW DPDT	SWITCH2	
1	TP1	SYNTH	HDR1x1	Test Point
1	TP2	FCLK	HDR1x1	Test Point
1	U1	MCP6H01(SOT23-5)	SOT23-5	Integrated Circuit
3	U2, U5, U11	MCP41010	SOIC8	Integrated Circuit
2	U3, U9	TSH82IYLT	SOIC8	Integrated Circuit
1	U4	LM4041	SOT23	Integrated Circuit
1	U6	PIC18F46K22-I/PT	QFP44	Integrated Circuit
1	U7	SN74VLC2G14	SOT23-6	Integrated Circuit
1	U8	LM7171	SOIC8	Integrated Circuit
1	U10	MAX9011	SOT23-6	Comparador
1	U12	AD9833	MICRO MSOP10	Integrated Circuit
1	U14	LM2576	TO263	Integrated Circuit
1	U15	LM79L05	SOIC8	
1	Y1	25MHz	XTAL SG7050	Oscillator



PCB-DDS-Rev3.PcbDoc San Jose